1 Planning (15 marks)

Defining the problem (3 marks)

P1 v is the independent variable and θ is the dependent variable or vary v and measure θ. [1]

P2 Keep the (shape and) size/volume/surface area/mass of balloon/helium constant [1]
Do not credit 'same balloon'.

P3 Keep the temperature (air/helium/balloon) constant. [1]

Methods of data collection (5 marks)

M1 Labelled diagram of apparatus: balloon, string fixed and method of producing wind.
Method of producing wind to be approximately horizontal to balloon. [1]

M2 Suspend mass from balloon. [1]

M3 Method to change wind speed, e.g. change setting, variable power supply/resistor/change distance from fan. [1]

M4 Method to measure wind speed, e.g. wind speed indicator/detector, anemometer [1]

M5 Method to measure angle – use protractor or rule for measurements for trigonometry methods.
This must be shown correctly on diagram or explained in text. [1]

Method of analysis (2 marks)

A1 Plot a graph of tan θ against $1/v^2$. [1]

A2 Relationship valid if straight line through origin [1]

Safety considerations (1 mark)

S1 Avoid the moving blades of the fan (safety screen, switch off when changing experiment); goggles to avoid air stream into eye. [1]

Additional detail (4 marks)

D1/2/3/4 Relevant points might include [4]

1 Large wind speed to produce measurable deflection/large cross-sectional area of balloon.
2 Additional detail on measuring angle e.g. use a large protractor, projection method.
3 tan $\theta = h/l$.
4 Measuring air speed at point where balloon is positioned.
5 Adjust height of fan so that air flow is horizontally aligned to the balloon.
6 Reason for adding mass to increase stability/deflection.
7 Keep windows shut/air conditioning switched off/use of wind tunnel to avoid draughts.
8 Wait for the balloon to become stable.

Do not allow vague computer methods.

[Total: 15]

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